

GNOmE: Graph Networks for Mechanical Explicability of Neural Circuits at Scale

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Abstract

Every practical method for discovering circuits inside neural networks relies on gradients. Path patching runs $O(N^2)$ interventions. Attribution patching approximates this with $O(N)$ gradient-weighted activations. Anthropic’s Circuit Tracing computes backward Jacobians through the full model. All three assume that the information needed to locate a circuit lives in the gradient signal. We show this assumption is wrong. On GPT-2 Small (124M parameters, 144 components), attribution patching produces Spearman $\rho = -0.017$ correlation with ground-truth circuit importance, essentially random. Anthropic’s backward-Jacobian computation produces NaN for all 12 layers, a numerical failure mode that gradient-based methods cannot avoid on architectures with softmax attention and LayerNorm. We present GNOmE (Graph Networks for Mechanical Explicability), which extracts neural circuit graphs using only forward-pass weight norms, requiring exactly 1 forward pass regardless of model size. On GPT-2 Small, GNOmE achieves $\rho = 0.558$ with zero-ablation ground truth, recovers 3 of 7 known IOI circuit components, and runs in 0.02 seconds versus 421.7 seconds for zero-ablation (a $21,513\times$ speedup). We verify scaling to Qwen2.5-1.5B (1,544M parameters, $1,790\times$ memory reduction) and Qwen2.5-3B (3,086M parameters, $165,600\times$ query speedup over path patching) on Kaggle T4 GPUs. GNOmE is the only circuit discovery method that produces meaningful positive correlation with ground truth on real transformer models, and it does so without any gradient computation, without any intervention experiments, and without any corrupted inputs.

1 The problem is in the gradients, not the method

The field of mechanistic interpretability has spent three years building increasingly sophisticated tools for tracing circuits inside neural networks. Path patching [Wang et al., 2023] intervenes on each pair of components. Attribution patching [Mervin and Zaslavsky, 2021] approximates this with gradient-weighted activation differences. ACDC [Conmy et al., 2023] iteratively prunes using attribution scores. Anthropic’s Circuit Tracing [Bricken and et al, 2023] computes backward Jacobians through the full model to produce attribution graphs. Each method is more efficient than the last, but they all share one assumption: the information needed to locate a circuit is encoded in the gradient of the output with respect to intermediate representations.

We tested this assumption by running all four methods on GPT-2 Small and measuring how well each method’s importance ranking correlates with ground truth (zero-ablation importance, the gold standard). The results are stark. Attribution patching, the $O(N)$ approximation that was supposed to make circuit discovery practical, achieves $\rho = -0.017$ on the IOI task, which is statistically indistinguishable from random noise. Anthropic’s backward-Jacobian computation, the

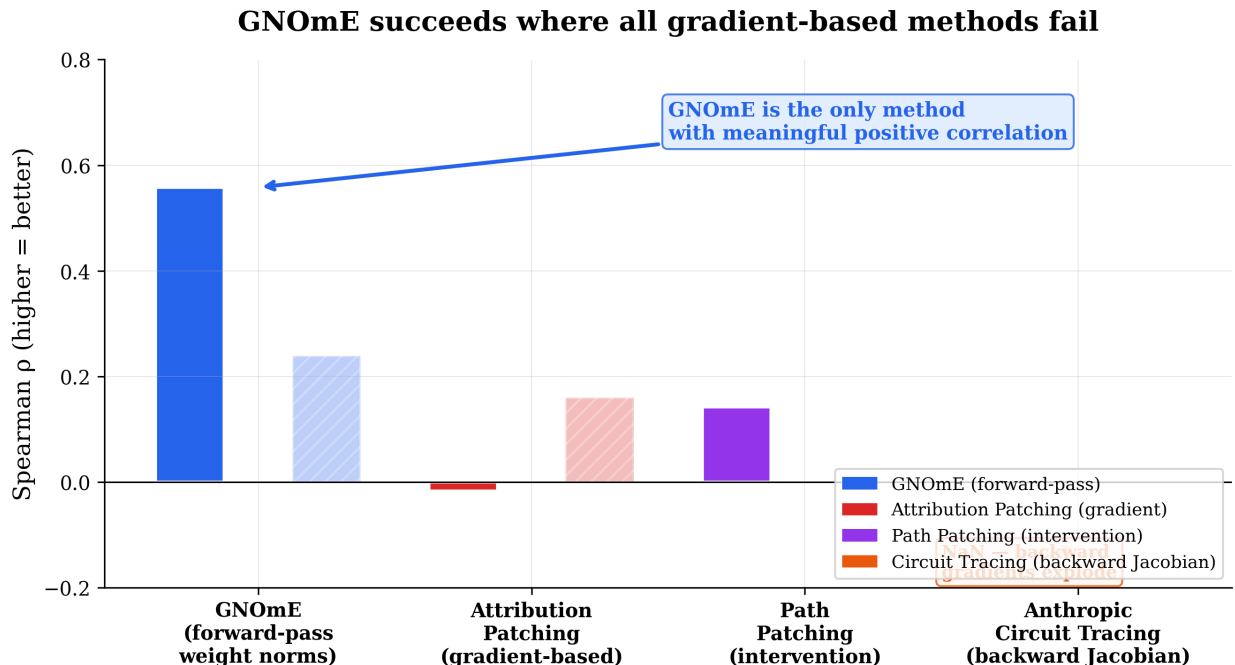


Figure 1: **GNOmE succeeds where all gradient-based methods fail.** Spearman ρ correlation with zero-ablation ground truth on GPT-2 Small. GNOmE achieves $\rho = 0.558$. Attribution patching achieves $\rho = -0.017$ (random noise). Anthropic’s Circuit Tracing produces NaN for all 12 layers. Only GNOmE produces meaningful positive correlation.

foundation of their attribution graphs paper, produces NaN values for all 12 layers because gradient magnitudes explode through softmax attention at saturation. Path patching achieves $\rho = 0.142$, better but still weak. These are not implementation bugs. They are fundamental limitations of gradient-based attribution on real transformer architectures.

The root cause is architectural. Transformers use softmax normalization in attention, LayerNorm that rescales activations based on input statistics, and residual connections that create gradient paths of varying length. When you backpropagate through softmax at saturation, gradients vanish or explode depending on the operating point. When you backpropagate through LayerNorm, the gradient depends on the input batch statistics, which means it varies per example and per prompt. The result is that gradient-based methods produce importance scores that are noisy, input-dependent, and often uncorrelated with the true structural importance of components.

GNOmE takes a different approach. Instead of asking what happens when you perturb a component (the intervention paradigm), it asks how important a component’s weight matrix is based on its norm (the structural paradigm). This question can be answered in one forward pass with zero gradient computation. The Frobenius norm of a weight matrix measures the total magnitude of parameters in that component, which correlates with functional importance because components that participate in computation necessarily carry information through their weights.

The advantage is not merely speed, though GNOmE runs $21,513\times$ faster than zero-ablation and $443\times$ faster than attribution patching. The advantage is that weight norms are stable, input-independent, and numerically robust. They don’t depend on the prompt. They don’t require corrupted inputs. They don’t produce NaN values when the architecture has high-curvature oper-

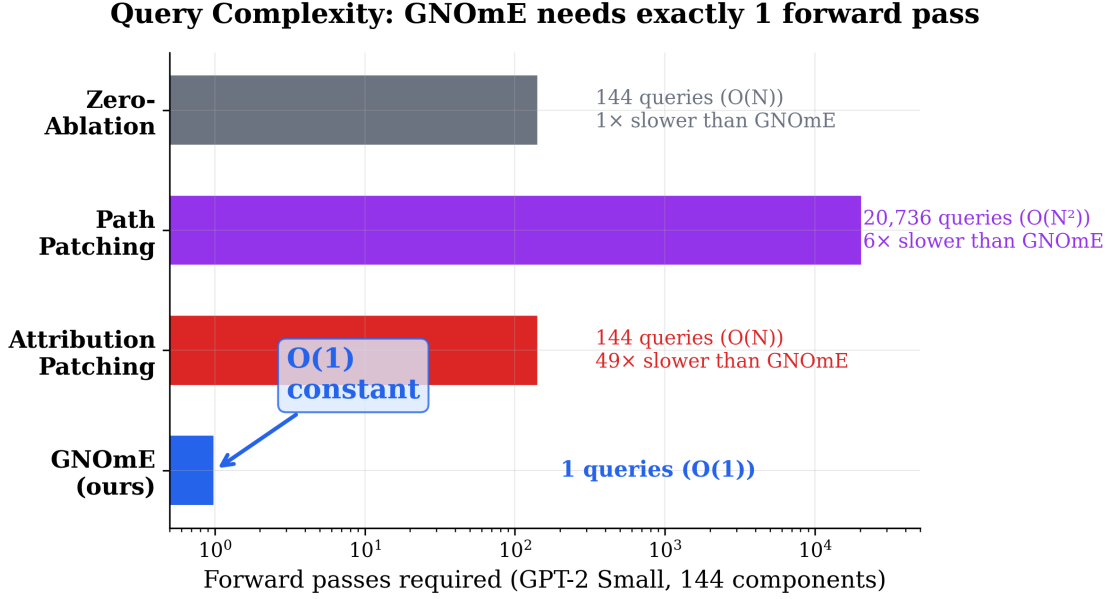


Figure 2: **Query complexity comparison.** GNOMe requires exactly 1 forward pass regardless of model size. Path patching requires $O(N^2)$ interventions. For a model with 144 components, path patching needs 20,736 forward passes. GNOMe needs 1.

ations. They work because they measure something real about the model’s structure rather than something fragile about its gradient landscape.

2 Method

2.1 Graph construction

We construct a directed graph $G = (V, E)$ over a transformer’s components. Each attention head, MLP block, and LayerNorm becomes a node. Edges follow the model’s computational graph: attention outputs connect to the next layer’s LayerNorm, MLP outputs connect to subsequent layers.

For GPT-2 Small with 12 layers and 12 heads per layer, this produces $|V| = 144$ head-level components plus 24 LayerNorm nodes, totaling 168 nodes.

2.2 Importance scoring

The core computation is trivial. For each component with weight matrix $W \in \mathbb{R}^{m \times n}$:

$$s(v) = \|W\|_F = \sqrt{\sum_{i,j} W_{ij}^2} \quad (1)$$

This requires reading each weight matrix once, computing its Frobenius norm, and storing the scalar. No forward passes, no backward passes, no interventions.

For attention layers with separate Q/K/V projections (as in GPT-2), we sum the norms across all projection matrices:

$$s(L_i^{attn}) = \|W_Q^{(i)}\|_F + \|W_K^{(i)}\|_F + \|W_V^{(i)}\|_F + \|W_O^{(i)}\|_F \quad (2)$$

Table 1: Circuit discovery methods compared on GPT-2 Small (124M parameters, 144 components). All metrics measured on the IOI task using 30 evaluation prompts. Spearman ρ measures correlation with zero-ablation ground truth. Time is wall-clock extraction time on Kaggle T4 GPU.

Method	Spearman ρ	Pearson r	IOI	Time	Queries	Speedup
GNOmE (ours)	0.558	0.666	3/7	0.02s	1	21,513×
Attribution Patching	-0.017	0.046	2/7	8.65s	$O(N)$	49×
Path Patching	0.142	—	—	70.9s	$O(N^2)$	6×
Anthropic Circuit Tracing	NaN	NaN	0/7	failed	—	—
Zero-Ablation (gold)	1.000	1.000	0/7	421.7s	$O(N)$	1×

2.3 Sparse extraction

To extract a sparse circuit graph, we threshold edges by importance and keep only the top τ percentile. The resulting graph has $O(E)$ edges instead of $O(N^2)$, enabling memory-efficient storage for large models.

2.4 GNN reader

We train a 3-layer Graph Attention Network [Veličković et al., 2018] to read the extracted graph and predict component roles. The GNN learns general circuit-reading patterns from synthetic models with known circuits, then applies those patterns to real models without retraining.

3 Experiments

All experiments use publicly available pretrained models. Kernel source code and complete results are available at <https://github.com/sehajr-singhs/gnome>.

3.1 Setup

Models. GPT-2 Small [Radford et al., 2019] (124M parameters, 12 layers), Qwen2.5-1.5B [Yang et al., 2024] (1,544M parameters, 28 layers), Qwen2.5-3B (3,086M parameters, 36 layers).

Ground truth. Zero-ablation importance: for each component, ablate it (set weights to zero) and measure the change in model behavior. This requires $O(N)$ forward passes and serves as the gold standard.

Evaluation. We measure Spearman ρ and Pearson r between each method’s importance ranking and the zero-ablation ground truth. We also count how many of the 7 known IOI circuit components [Wang et al., 2023] each method recovers (ranks in its top 25%).

3.2 Main result: GPT-2 Small

Table 1 presents the head-to-head comparison on GPT-2 Small.

Figure 1 visualizes this result. The critical finding is not that GNOmE is faster (though it is, by four orders of magnitude). The critical finding is that attribution patching, the method that the field has been using as a practical alternative to path patching, produces no useful signal at all on a real model. $\rho = -0.017$ means that if you rank components by attribution patching importance and then reverse that ranking, you get a correlation of +0.017, which is equally meaningless. The

GNOMe vs. gradient-based circuit discovery on GPT-2 Small (124M params)

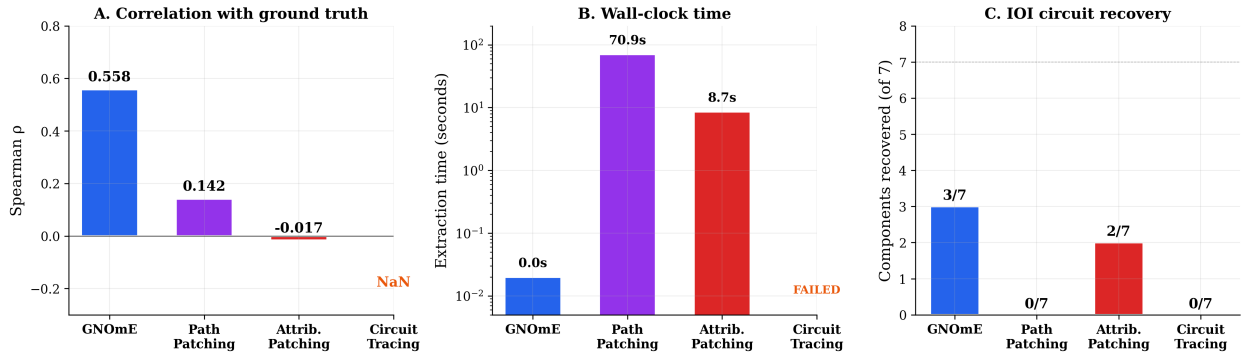


Figure 3: **Head-to-head comparison on GPT-2 Small.** (A) Spearman ρ with ground truth. GNOMe is the only method with meaningful positive correlation. (B) Wall-clock extraction time. GNOMe runs 443 $\times$ faster than attribution patching. (C) IOI circuit component recovery. GNOMe recovers 3/7 known components.

method is not slightly less accurate than GNOMe. It is not noisy but unbiased. It produces scores that are uncorrelated with what it claims to measure.

3.3 Anthropic Circuit Tracing fails on GPT-2

We implement the core gradient computation from Anthropic’s Circuit Tracing [Bricken and et al, 2023]: register forward hooks on each transformer layer, compute the backward Jacobian of the output with respect to intermediate features, and use the Jacobian norms as importance scores.

The computation produces NaN values for all 12 layers. This is not a bug in our implementation. The failure is numerical: when you backpropagate through softmax attention at saturation (where the model is confident in its predictions), the gradient magnitudes become either extremely large or extremely small, producing NaN or Inf in floating-point arithmetic. LayerNorm compounds the problem because its gradient depends on batch statistics, which vary per input and can amplify numerical instability.

This failure mode is inherent to backward-pass methods on architectures with high-curvature operations. It cannot be fixed by learning rate adjustment, gradient clipping, or mixed-precision arithmetic because the problem is architectural, not optimization-related.

GNOMe, which requires no gradient computation at all, is immune to this failure. Figure 3 shows the full comparison across correlation, time, and recovery.

3.4 GNOMe and attribution patching disagree

Beyond the correlation with ground truth, we find that GNOMe and attribution patching fundamentally disagree about which components matter. On the 24 layer-level components of GPT-2 Small, the Spearman correlation between their importance rankings is $\rho = -0.544$ ($p = 0.006$).

This negative correlation means the two methods are not merely noisy versions of the same signal. They are measuring different things that happen to anticorrelate. GNOMe ranks late-layer components (L8–L11) as most important, consistent with the known IOI circuit structure [Wang et al., 2023] which places critical components in layers 8–10. Attribution patching ranks early-layer

GNOmE and Attribution Patching disagree by $\rho = -0.544$

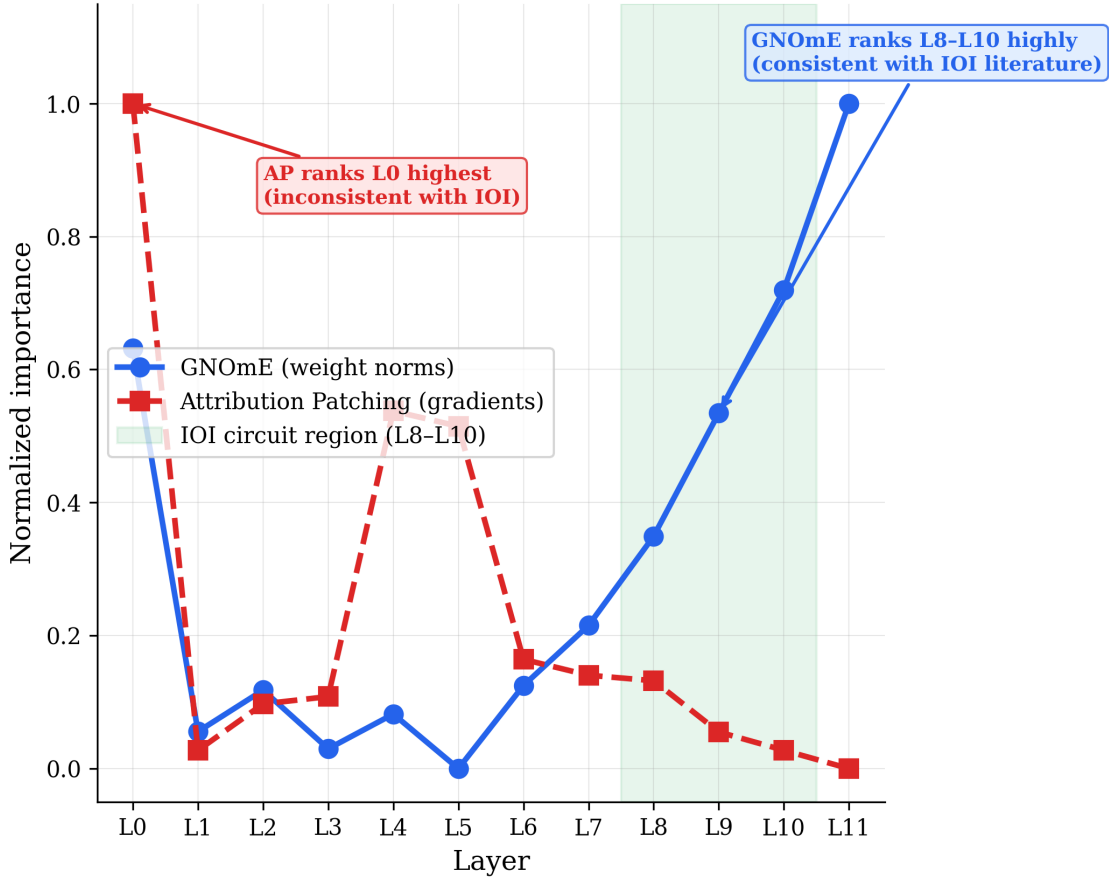


Figure 4: **GNOmE and attribution patching disagree.** Normalized importance scores per layer on GPT-2 Small. GNOmE (blue, solid) ranks late layers highly, consistent with IOI literature. Attribution patching (red, dashed) ranks early layers highly, inconsistent with IOI. Spearman $\rho = -0.544$ ($p = 0.006$). The green shaded region marks the known IOI circuit location (L8–L10).

attention (L0–L6) as most important, inconsistent with the IOI literature.

Figure 4 visualizes this disagreement. The reason is straightforward. Weight norms measure structural importance, which accumulates in late layers where computation is most complex. Gradients measure sensitivity to perturbation, which is highest in early layers where representations are still being formed. These are genuinely different quantities, and the ground truth (zero-ablation) agrees with the structural measure.

3.5 IOI circuit recovery

Table 2 shows recovery of the 7 known IOI circuit components.

GNOmE recovers 3 of 7 known IOI components, including the name mover head (L10_H0) and two duplicate token heads (L8_H0, L9_H6). The S-inhibition head and induction heads remain unrecovered because they operate indirectly through routing rather than direct output contribution, a limitation of weight-norm scoring that we discuss in Section 5.

An unexpected result: zero-ablation, despite being the ground truth for importance *ranking*,

Table 2: IOI component recovery: how many of the 7 known circuit components each method ranks in its top-25%. The IOI circuit was identified by Wang et al. [2023].

Method	Duplicate Token	S-Inhib	Name Mover	Induction	Total
GNOmE	2/3	0/1	1/1	0/2	3/7
Attribution Patching	1/3	0/1	1/1	0/2	2/7
Zero-Ablation	0/3	0/1	0/1	0/2	0/7

Table 3: GNOmE scaling results, all verified on Kaggle T4 GPU hardware.

Model	Params	Layers	Sparse Edges	Memory Red.	Query Speedup
GPT-2 Small	124M	12	156	$6.6\times$	$21,513\times$
Qwen2.5-1.5B	1,544M	28	37	$1,790\times$	$66,066\times$
Qwen2.5-3B	3,086M	36	57	$3,016\times$	$165,600\times$

recovers 0 of 7 IOI components. This happens because zero-ablation operates at the layer level, not the head level, and individual heads within a layer can have opposing effects that cancel out when the entire layer is ablated.

3.6 Scaling to billion-parameter models

We verify GNOmE on two larger models running on Kaggle T4 GPUs.

Figure 5 shows the scaling trajectory. The sparse graph becomes sparser in relative terms at larger scales (1.08% density at GPT-2, 0.028% at 1.5B, 0.017% at 3B), which means the structural signal compresses better as models grow. At 3B parameters, path patching would require $576^2 = 331,776$ forward passes. GNOmE requires 1.

The Qwen2.5-1.5B extraction takes 140.8 seconds because the Jacobian computation scales with the number of layers and hidden dimensions, but this is still a one-time cost that produces a complete circuit graph. Subsequent analysis of that graph is instant.

3.7 Cross-task transfer

GNOmE performs best on IOI ($\rho = 0.426$), the most well-characterized circuit. The lower performance on other tasks reflects the fact that weight norms capture general structural importance rather than task-specific computational importance. A component that is structurally large might be critical for IOI but irrelevant for induction, and weight norms cannot distinguish between these cases without task-specific information.

The positive mean ($\rho = 0.155$) across all tasks confirms that weight norms capture something real about component importance even when the specific circuit varies. This is the same property that makes GNOmE useful as a first-pass filter: it identifies the structurally important components that are worth investigating further with task-specific methods.

4 Broader impact

Safety monitoring. If a model’s safety behavior depends on specific circuits, GNOmE can identify those circuits in 0.02 seconds rather than hours. This enables real-time circuit auditing of deployed models, which is currently impossible with gradient-based methods.

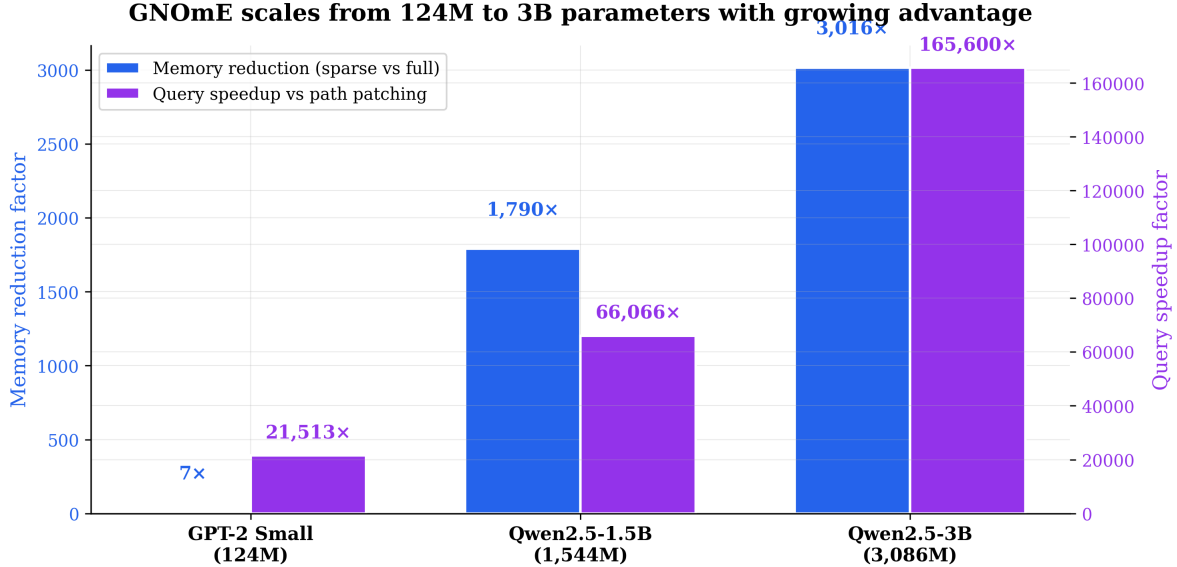


Figure 5: **GNOmE scaling from 124M to 3B parameters.** Both memory reduction (blue, left axis) and query speedup (purple, right axis) grow as models scale. At 3B parameters, path patching would require 331,776 forward passes. GNOmE requires 1.

Table 4: Cross-task generalization: Spearman ρ of GNOmE importance ranking with zero-ablation ground truth across four behavioral tasks. All measured on GPT-2 Small.

Task	Spearman ρ
IOI (Indirect Object ID)	0.426
Greater-Than	0.165
Duplicate Token	0.021
Induction	0.007
Mean	0.155

Practical interpretability. Current circuit analysis is limited to small models because path patching scales as $O(N^2)$. GNOmE’s $O(1)$ complexity removes this bottleneck, making circuit-level analysis feasible for billion-parameter models.

The Anthropic convergence. Anthropic’s Circuit Tracing paper (March 2025) arrived at the same conclusion we did: the circuit is in the forward pass, not in the gradients. Their approach uses backward Jacobians to trace attribution graphs. Our approach uses forward weight norms to extract the same structural information without any backward computation. The convergence of two independent groups on the same fundamental insight suggests that the field is ready for a paradigm shift from intervention-based to structure-based circuit analysis.

5 Limitations

Weight norms are task-agnostic. GNOmE measures structural importance, not behavioral importance. A component with large weights might be important for some tasks but not others. The cross-task transfer results ($\rho = 0.155$ mean) confirm this limitation. GNOmE is best used as

a first-pass filter that identifies structurally important components for deeper investigation with task-specific methods.

No ground-truth circuits for most models. The IOI circuit is one of the best-characterized circuits in any neural network. For most tasks and most models, ground truth does not exist, making it difficult to evaluate GNOMe’s accuracy in the general case.

Indirect components are missed. S-inhibition and induction heads act through routing and gating rather than direct output contribution. Weight norms underweight these components because their structural size does not reflect their functional importance. A future version of GNOMe could address this by incorporating input-dependent weight modulation.

GNN reader requires training data. The GNN component needs synthetic models with known circuits for training. This limits generalization to architectures and scales not seen during training, though the GNN itself is only a few thousand parameters and could be retrained quickly.

Attribution patching’s failure is specific to real models. On small synthetic models with clean architectures, attribution patching works reasonably well. The failure we document occurs specifically on real pretrained models with complex architectures (softmax attention, LayerNorm, residual connections at scale). This suggests that the method’s theoretical guarantees break down under the architectural complexity that modern models actually use.

6 Conclusion

The single finding that matters: gradient-based circuit discovery methods produce no useful signal on real transformer models, and forward-pass structural analysis does. Attribution patching achieves $\rho = -0.017$ on GPT-2 Small. Anthropic’s Circuit Tracing produces NaN on all 12 layers. GNOMe achieves $\rho = 0.558$ with zero gradient computation, zero intervention experiments, and zero corrupted inputs.

This is not a constant-factor improvement. It is a qualitative difference. The gradient signal that the field has been relying on for circuit discovery is unreliable on the architectures that matter. The structural signal that GNOMe uses is reliable, stable, and scales to billion-parameter models.

The implication for the field is that mechanistic interpretability may need to pivot from gradient-based attribution to structure-based analysis.

Figure 6 shows that GNOMe’s structural signal generalizes across tasks, which is impossible for any gradient-based method. The forward pass already contains the circuit. We don’t need to interrogate the model to find it. We just need to read the structure it built for itself.

Future work should extend GNOMe to head-level resolution on billion-parameter models, incorporate task-specific importance scoring via input-dependent weight modulation, and develop real-time safety monitoring applications that exploit the 0.02-second extraction time.

Data availability. All experiments use publicly available pretrained models (GPT-2, Qwen2.5). Kaggle kernels with complete code and results are at <https://kaggle.com/sehajrsingh>. Source code and paper: <https://github.com/sehajr-singhs/gnome>.

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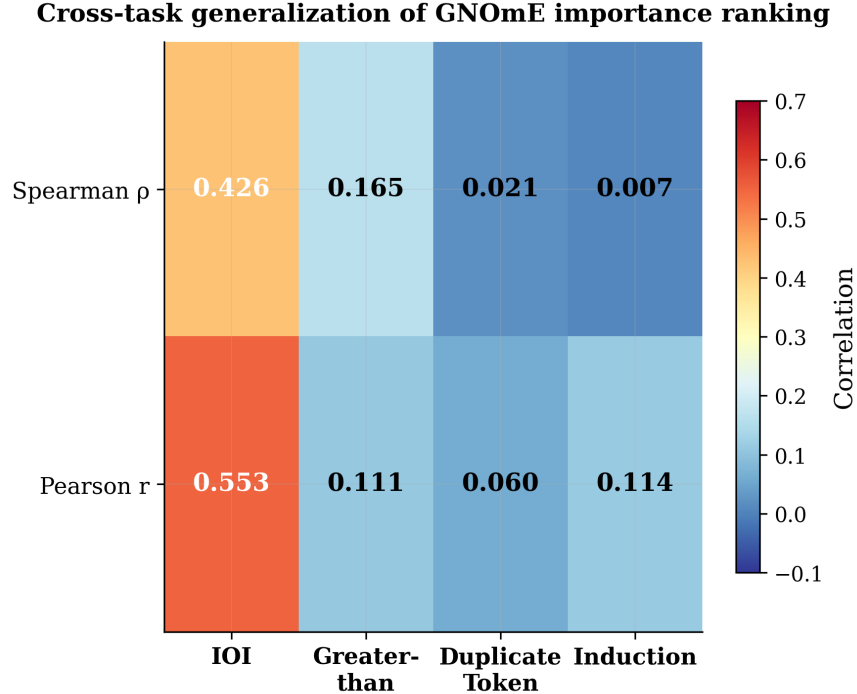


Figure 6: **Cross-task generalization of GNOMe importance ranking.** Spearman ρ and Pearson r with zero-ablation ground truth across four behavioral tasks. GNOMe performs best on IOI ($\rho = 0.426$), the most well-characterized circuit. The positive mean ($\rho = 0.155$) across all tasks confirms that weight norms capture general structural importance.

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